

MovableBag: Substitutional Robot for Enhancing Immersive Boxing Training with Encountered-Type Haptic

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ABSTRACT

Boxing is a combat sport that involves fierce movement and close body contact between two boxers. In addition to the boxer, the coach is also essential for training. The coach could be injured or too tired to continue consecutive training sessions for many boxers. Thus, we present MovableBag, a substitutional robot that provides room-scale mobility and the ability to carry sport props with an easier tracking setup to simulate the boxing opponent or coach in virtual reality with encountered-type haptic feedback. In addition, we integrate our system with a wireless VR headset and develop a VR boxing application to show the capacity of the device and inform future studies.

CCS CONCEPTS

• Human-centered computing → Haptic devices; • Hardware → Emerging interfaces.

KEYWORDS

Immersive Training, Substitutional Robot, Encountered-Type Haptic, Virtual Reality, Boxing

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1 INTRODUCTION

To improve physical and psychological training, many studies have been conducted to investigate the application of mixed and augmented reality technology in the sports industry, particularly in training for professionals and amateurs, including mental training,

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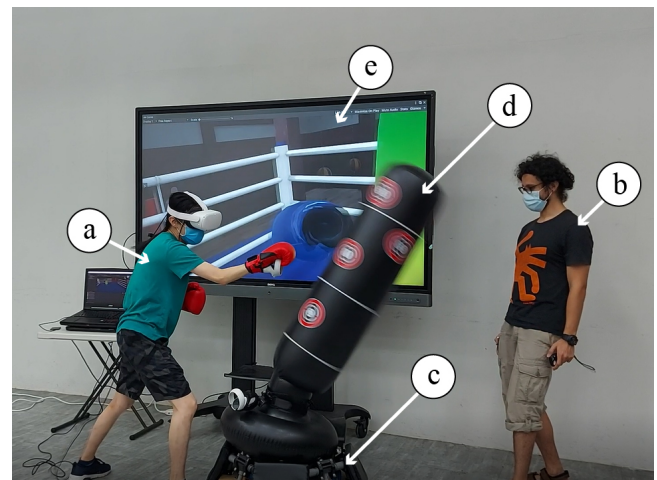


Figure 1: VR Boxing Training with MovableBag: (a)Boxer, (b)Facilitator(Coach), (c)MovableBag(Base), (d)Target props(Punching airbag), (e) Boxer's view in VR headset

injury prevention, and performance monitoring.[Sawan et al. 2020] In combat sports, haptic feedback plays a critical role in muscle training, such as punching.

Encountered-type haptics have been investigated in physical environments to provide realistic haptic sensations to users. Most devices offer mobility, but subtle haptic feedback is provided using small swarm robots and props. For example, REACH+[Gonzalez et al. 2020] and HapticBots[Suzuki et al. 2021] offer haptics to the fingertips or a small area of the hand; Tangiball[Bozgeyikli and Bozgeyikli 2022] uses a spherical structure containing a tracker for foot-enabled haptic interactions in VR; RoomShift[Suzuki et al. 2020] uses a swarm robot to move lightweight furniture.

Regarding VR application in sports training with encountered-type haptic, most of the works are currently designed to be used at a fixed spot to withstand strong impact and heavy weight without mobility. InfinityWall [Kosmalla et al. 2022] shows a rock climbing treadmill for long-distance vertical climbing in VR at a fixed physical location. VCoach[Chen et al. 2022] uses a wearable tracker to

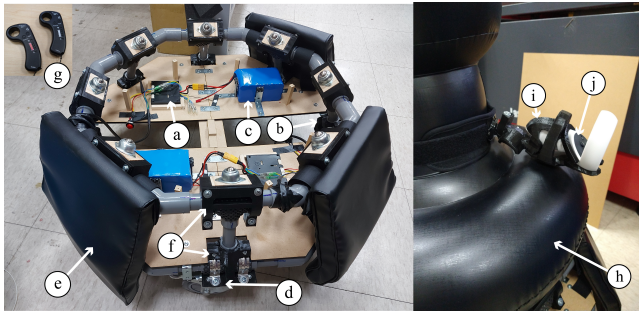


Figure 2: Structure Overview of MovableBag: (a)Wheel hub motor driver, (b)Motor wheel, (c)Li-ion battery, (d)Omnidirectional wheel, (e)Cushions, (f)3D-printed connectors, (g)Driver controller, (h)Target prop, (i)3D-printed holder, (j)Oculus Touch Controller

monitor punching motion that can withstand high physical impact, but the device itself does not provide mobility for agility training.

Therefore, it would be a new approach to enhance both device mobility and the strength of encountered-type haptics in the VR application. CoVR[Bouzbib et al. 2020] shows strong kinesthetic feedback for pushing and transporting heavy props, but only along ceiling tracks without rotation flexibility and also requires a complex setup. In this paper, we design a robot with greater mobility and using a simpler setup for tracking, allowing for more fierce movement in the VR environment. For example, strength and agility training in sports. Additionally, a VR boxing application has been developed to show the utility of our robot.

2 DESIGN AND IMPLEMENTATION

We present MovableBag, a substitutional robot that incorporates encountered-type haptic to enhance the VR experience. Our robot consists of (a) a movable base and (b) target props. The Movable base is designed for providing mobility and bear loading for props, which utilizes several components: (i) wheel motor drivers, controlling the speed and direction of two motor wheels, (ii) two omnidirectional passive wheels to allow robot support stronger impact and prop, and (iii) cushions surrounding the base for users' safety and damage reduction to the robot due to crash. The movable base is made of plastic pipes, and we designed 3D-printed connectors to build the octagonal structure. The mobile base weighs 14.3 kg with a maximum speed of 18 km/h and maximum loading of approximately 70kg according to the specification of the hub motor drivers (24V). Additionally, we designed a 3D-printed holder for Oculus Touch Controller attached on the target prop (punching airbag, weighted 6.6kg) for location tracking and haptic reaction in VR application.

3 APPLICATION: VR BOXING TRAINING

We use Unity to design a virtual environment(VE) for boxing training to demonstrate our design concept. One user is the coach(facilitator) who controls the movement of the MovableBag. Another user is the boxer who wears the boxing glove with another VR controller, which is represented as a virtual glove in the VE to provide immersiveness to users and allow punching interaction. The physical

location and movement of the target prop is constantly monitored by the attached VR controller with its internal accelerometer and gyroscope sensors. The virtual airbag is then visualized in VE based on the sensor data stream. The coach can control the robot and give instructions to the boxer to train the punching with boxing foot-work. Whenever the boxer punches the target prop, the boxer can receive direct haptic feedback from the punching bag in the physical environment. The sensor data from the VR controller attached on the target prop will be calculated to generate visual feedback for the virtual bag in VE.

4 DISCUSSION AND FUTURE WORK

This work presents a substitutional robot to enhance immersive experience with encountered-type haptic by carrying a prop for boxing training with greater mobility on a room scale. We also present a simple tracking setup with the robot to enable more fierce movement and encountered-type haptic feedback from punching to simulate the coach for boxing training in the VE. In the future, the feature of capturing physical data, such as punch strength, will be developed for performance monitoring. Additionally, we will adopt participatory design by inviting coaches into the design process. It helps to investigate more design considerations to improve its training effectiveness and to explore more applications of this work in different combat sports, such as Karate and Taekwondo.

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REFERENCES

- Elodie Bouzbib, Gilles Bailly, Sinan Haliyo, and Pascal Frey. 2020. CoVR: A Large-Scale Force-Feedback Robotic Interface for Non-Deterministic Scenarios in VR. Association for Computing Machinery, New York, NY, USA, 209–222. <https://doi.org/10.1145/3379337.3415891>
- Lai Lila Bozgeyikli and Evren Bozgeyikli. 2022. Tangiball: Foot-Enabled Embodied Tangible Interaction with a Ball in Virtual Reality. In *2022 IEEE Conference on Virtual Reality and 3D User Interfaces (VR)*. IEEE, 812–820.
- Hao Chen, Yujia Wang, and Wei Liang. 2022. VCoach: Enabling Personalized Boxing Training in Virtual Reality. In *2022 IEEE Conference on Virtual Reality and 3D User Interfaces Abstracts and Workshops (VRW)*. IEEE, 546–547.
- Eric J. Gonzalez, Parastoo Abtahi, and Sean Follmer. 2020. REACH+: Extending the Reachability of Encountered-Type Haptics Devices through Dynamic Redirection in VR. In *Proceedings of the 33rd Annual ACM Symposium on User Interface Software and Technology (Virtual Event, USA) (UIST '20)*. Association for Computing Machinery, New York, NY, USA, 236–248. <https://doi.org/10.1145/3379337.3415870>
- Felix Kosmalla, Florian Daiber, and Antonio Krüger. 2022. InfinityWall – Vertical Locomotion in Virtual Reality Using a Rock Climbing Treadmill. In *CHI Conference on Human Factors in Computing Systems Extended Abstracts* (New Orleans, LA, USA) (CHI EA '22). Association for Computing Machinery, New York, NY, USA, Article 280, 6 pages. <https://doi.org/10.1145/3491101.3519654>
- Nedal Sawan, Ahmed Eltwari, Caterina De Lucia, Luigi Pio Leonardo Cavaliere, Alessio Faccia, and Narcisa Roxana Moşteanu. 2020. Mixed and augmented reality applications in the sport industry. In *2020 2nd International Conference on E-Business and E-commerce Engineering*. 55–59.
- Ryo Suzuki, Hooman Hedayati, Clement Zheng, James L Bohn, Daniel Szafir, Ellen Yi-Luen Do, Mark D Gross, and Daniel Leithinger. 2020. Roomshift: Room-scale dynamic haptics for vr with furniture-moving swarm robots. In *Proceedings of the 2020 CHI conference on human factors in computing systems*. 1–11.
- Ryo Suzuki, Eyal Ofek, Mike Sinclair, Daniel Leithinger, and Mar Gonzalez-Franco. 2021. HapticBots: Distributed Encountered-Type Haptics for VR with Multiple Shape-Changing Mobile Robots. In *The 34th Annual ACM Symposium on User Interface Software and Technology (Virtual Event, USA) (UIST '21)*. Association for Computing Machinery, New York, NY, USA, 1269–1281. <https://doi.org/10.1145/3472749.3474821>